

# A Review of Power System Challenges Stemming from Large AI Data Center Loads

Venkateswara R. Seshmasetti,  
Department of Electrical Engineering  
& Computing Sciences  
Rochester Institute of Technology  
Dubai, United Arab Emirates (UAE)  
[vrs6329@rit.edu](mailto:vrs6329@rit.edu)

Abdulla Ismail  
Department of Electrical Engineering  
& Computing Sciences  
Rochester Institute of Technology  
Dubai, UAE  
[axicad@rit.edu](mailto:axicad@rit.edu)

Haris M. Khalid,  
College of Engineering & IT  
University of Dubai, Dubai, UAE  
Department of Electrical and  
Electronic Engineering Science  
University of Johannesburg,  
Auckland Park, South Africa  
[mkhaliid@ud.ac.ac](mailto:mkhaliid@ud.ac.ac)

**Abstract**—The deployment of Artificial Intelligence (AI) data centers presents significant challenges for electrical power systems. This proposed paper reviews the impacts of AI data center loads on power system operation by systematically analyzing technical documents, utility reports, and regulatory frameworks. Unlike traditional loads, AI data centers exhibit distinct power dynamics, independent protection philosophies, and geographic concentration, which can destabilize both local and system levels. Notable incidents, such as sharp power fluctuations and load disconnections, raise concerns about system stability. This proposed review aims to synthesize knowledge on this emerging topic by developing a multi-timescale framework, validating theoretical predictions, classifying AI-induced oscillatory modes (0.5-23 Hz), examining stability effects in Inverter-Based Resources (IBR) systems, and proposing hierarchical mitigation strategies. These insights provide vital guidance for operators and researchers on the impacts of AI data centers on the grid.

**Index Terms**—AI data centers, inverter-based resources, large loads, multi-timescale dynamics, oscillatory phenomena.

## I. INTRODUCTION

The transformation of utility power systems due to increased AI data center loads marks a significant shift in power system evolution. Since the November 2022 public deployment of ChatGPT, the infrastructure supporting these AI computations has expanded rapidly. The existing literature indicates that AI data centers are distinct from traditional computing loads, referred to as “Large Loads,” [1] “programmable loads,” [2] or “negative generators,” [3] and introduce unpredictable load variations. Data from Google’s AI data center shows that these fluctuations can reach tens of megawatts within sub-second intervals [4][5], while affecting critical system oscillation modes.

This proposed review highlights the concerning convergence of growing AI data center loads, which are primarily power electronic loads, with inverter-based resources (IBR) dominating power generation. This pairing reduces system synchronous inertia and negatively impacts damping capability [3]. The July 2024 incident in the Eastern Interconnection, where 1,500 MW of data center load disconnected simultaneously, exemplifies the vulnerability to cascading failures [6]. The incident revealed oscillations at 14.7 Hz for 2 hours, indicating a new phenomenon outside traditional frequency ranges [7].

Technical analysis shows distinct differences between AI data centers and traditional computing loads, underscoring the need for a deeper examination of their effects on power systems. Operational data from Google indicates power fluctuations reaching tens of megawatts in sub-second intervals, a behavior unseen in conventional loads. The Eastern Interconnection incident exposes the limitations of existing analytical frameworks in addressing cascade failures.

This review faced challenges due to the novelty of the topic, with significant research emerging only since 2023 [1]. Comprehensive references were scarce, and currently, only the Alberta Electric System Operator (AESO) has published interconnection guidelines [8]. The Electric Reliability Council of Texas (ERCOT) has shared valuable operational data, while the North American Electric Reliability Corporation (NERC) is actively addressing these challenges with utilities and researchers. This synthesis offers insights that validate theoretical predictions through field events.

## A. Motivation of the Proposed Work

The rapid growth of AI data centers since the November 2022 rollout of ChatGPT is expected to introduce significant challenges for existing power system infrastructure. Recent incidents, such as a simultaneous disconnection of 1,500 MW in the Eastern Interconnection in July 2024 and extended 14.7 Hz oscillations in the Dominion Energy system, highlight these issues. These events indicate a shift from traditional power system behavior, with increasing IBR penetration exacerbating stability risks. Limited technical understanding in existing literature, combined with the urgency of these challenges, drives the need for a comprehensive review.

## B. Scope and Potential of the Work

The proposed paper examines the impacts of AI data centers across temporal scales, ranging from microsecond power electronics interactions to hourly thermal cycling patterns. The analysis encompasses 1) technical phenomena and their interconnections, synthesis of real operational incidents from multiple utilities, 2) evaluation of emerging mitigation technologies, and 3) assessment of developing regulatory frameworks.

This comprehensive review is expected to provide essential guidance for power system planners, operators, regulatory bodies, and researchers who must manage the challenges arising from exponentially growing AI computational demands.

## C. Preceding Affined Reviews and Main Contribution of the Proposed Work

This proposed paper advances the current understanding through the following contributions that address critical gaps in existing literature (See **Fig. I** for the graphical abstract):

- **Multi-timescale Framework:** Development of a comprehensive analysis demonstrating the cascade from microsecond switching transients through millisecond protection operations and second-scale ramping to hourly thermal patterns, revealing critical interdependencies absent from single-scale studies.
- **Operational Validation:** Integration of field incidents from Eastern Interconnection, Dominion Energy, and

ERCOT with theoretical analysis, providing empirical validation of previously theoretical phenomena.

- **Oscillation Taxonomy:** Systematic classification of AI-induced oscillatory phenomena spanning 0.5-23 Hz, including detection methodologies and mitigation approaches.
- **IBR-AI Data Center Interaction Analysis:** Review revealing compound stability challenges in IBR-dominated grids, synthesizing findings that demonstrate fundamental behavioral changes at high IBR penetration levels, particularly the critical 70% threshold identified in recent studies where traditional mitigation approaches fail.
- **Hierarchical Mitigation:** Formulation of time-matched solutions ranging from microsecond supercapacitor response to hourly thermal storage, integrated with regulatory frameworks.

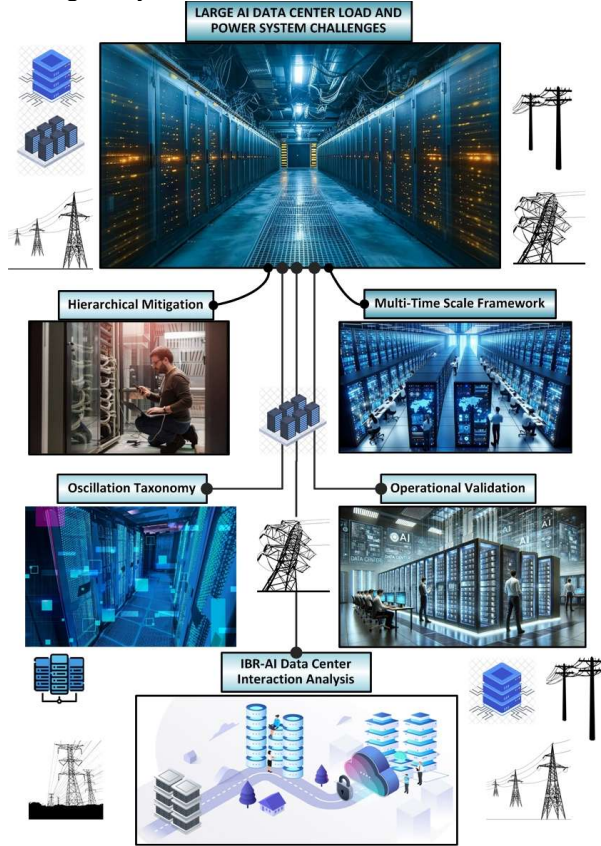


Figure 1. Graphical abstract of the proposed scheme.

**Table I** represents the preceding affirmed reviews in this area, the key limitations, and possible contributions addressed through the proposed work.

TABLE I. PRECEDING AFFIRMED REVIEWS AND CONTRIBUTIONS OF THIS WORK<sup>1</sup>

| Ref. | Focus Area                    | Key Limitations                                      | Addressed Work                                                           |
|------|-------------------------------|------------------------------------------------------|--------------------------------------------------------------------------|
| [1]  | Large load characteristics    | No multi-timescale technical analysis                | Multi-timescale framework                                                |
| [2]  | Programmable load flexibility | Single-timescale focus, lacks operational validation | Validates with real incidents, covers all timescales                     |
| [3]  | AI load ramping effects       | Restricted to frequency stability                    | Review stability assessment, including voltage and oscillatory phenomena |

|     |                        |                              |                                                             |
|-----|------------------------|------------------------------|-------------------------------------------------------------|
| [4] | Wide area oscillations | Examines a single phenomenon | Comprehensive oscillation taxonomy with interaction effects |
| [5] | Data center modeling   | No operational validation    | Review documented system operational incidents              |

1. AI is the acronym of artificial intelligence.

## II. AI DATA CENTER DEMAND GROWTH AND CHARACTERISTICS

This section comprises 1) quantitative growth trajectories and uncertainty, 2) geographic concentration and consequences, and AI workload power signatures.

### A. Quantitative Growth Trajectories and Uncertainty

The projection reports indicate significant uncertainty in AI data center growth. The International Energy Agency estimates global data center consumption will rise from 460 TWh in 2022 to over 1000 TWh by 2026 [9]. In the U.S., projections vary widely; IEEE PES anticipates 6.7-12% of total electricity usage by 2028 [35], while the Lawrence Berkeley National Laboratory forecasts 325 to 580 TWh [10]. A review by Kamiya and Coroam̃a suggests global energy use at 300-380 TWh in 2023, with AI-specific consumption potentially reaching 35-50% by 2030 [32]. EPRI estimates range from 196 to 404 TWh by 2030 [11]. The variation in projections stems from AI efficiency improvements and infrastructure limitations [31]. For instance, training GPT-4 consumed 62,318 MWh over 100 days [11], while inference operations show a significant increase in energy use per query compared to traditional searches [12][13]. Given billions of queries daily, this leads to a substantial increase in power grid demand.

### B. Geographic Concentration and Consequences

The geographic distribution of AI data centers shows unique patterns, influenced significantly by transmission capacity and cooling resource availability. These factors may lead to extreme concentration, creating vulnerabilities that extend beyond mere capacity issues. Research highlights the risk of cascade failures when clustered facilities react simultaneously to grid disturbances [5][14]. The July 2024 Eastern Interconnection event clearly demonstrated this, as multiple substations disconnected within seconds [6].

### C. AI Workload Power Signatures

Technical analysis reveals distinct power consumption patterns in AI workloads, which impact the grid differently. AI Training Workloads exhibit sustained high power with fluctuations, modeled as pulse trains with 60-80% duty cycles [4][15]. Google's operational data reveals significant power swings due to alternating computer and networking phases, with forcing frequencies of 0.5-1.5 Hz overlapping with traditional power system oscillation modes, posing potential risks [5]. Fine-tuning AI operations exhibit higher-frequency variation (0.3-0.7 Hz) and lower-amplitude power variations. Inference workloads generate burst patterns with lower average power but higher instantaneous ramp rates. Overall, computational resources scale dynamically with demand [15][16]. Measurements by Latif et al. on 8-GPU H100 nodes show 8.4 kW max power draw versus a manufacturer rating of 10.2 kW, indicating capacity planning discrepancies [33].

## III. POWER SYSTEM CHALLENGES: AN INTEGRATED ANALYSIS

This section comprises 1) multi-time scale channel propagation, 2) oscillatory phenomenon and resonance mechanisms, 3) voltage sensitivity and protection

miscoordination, and 4) IBR integration compound effects, respectively.

#### A. Multi-Time Scale Channel Propagation

The investigation highlights that AI data center challenges span multiple time scales, with each influencing the others. This creates a cascade effect from microsecond-level power electronics switching to hour-scale thermal cycling. Key phenomena, their grid impacts, measurement needs, and references are summarized in **Table II**. At the microsecond scale, power electronics in servers cause switching transients, with Li and Li identifying bandwidth limitations [17]. Upstream AC/DC converters limit system response, resulting in a 30-50 millisecond lag between GPU load changes and grid-visible power variations. These microsecond phenomena lead to millisecond-scale protection challenges, where voltage disturbances lasting 42-66 milliseconds can trigger data center protection [6]. The interaction between utility reclosing schemes and the "three strikes" disconnection logic complicates these issues [18][19]. At the second-to-minute scale, ramping poses significant operational challenges due to software-driven load variations. NERC documents indicate ramp rates of 1.9 p.u. per second [1], with the xAI facility showing 10-20 MW variations multiple times per second [20], exceeding traditional generation capabilities and posing serious frequency regulation issues [34].

TABLE II. MULTI-TIMESCALE CHALLENGE<sup>2</sup>

| Ref.                             | Time Scale | Primary Phenomenon                  |
|----------------------------------|------------|-------------------------------------|
| [17] [21]                        | $\mu$ s    | PE switching                        |
| [6][7][8]                        | ms         | Protection operations               |
| [4][5][20][32]                   | sec        | AI workload transitions             |
| [16][22]                         | min        | Thermal management cycling          |
| [15][23]                         | H          | Training epoch patterns             |
| <b>Grid Impact</b>               |            | <b>Measurement Challenge</b>        |
| Harmonic generation              |            | Beyond SCADA capability             |
| Voltage-triggered disconnections |            | Requires PMU/DFR equipment          |
| Frequency deviations             |            | Standard SCADA sufficient           |
| Reactive power variations        |            | Correlation with temperature        |
| Sustained oscillations           |            | Pattern recognition/source location |

<sup>2</sup> DFR, PE, PMU, and SCADA are the acronyms of digital fault recorder, power electronics, phasor measurement unit, and supervisory control and data acquisition, respectively.

#### B. Oscillatory Phenomenon and Resonance Mechanisms

AI data centers are likely to introduce new oscillatory phenomena in power systems, which will be a significant concern for future operations. Sub-synchronous oscillations, such as 14.7 Hz from UPS instability [7] and 23 Hz from large loads [24], pose risks through torsional interactions that standard SCADA systems fail to detect. These high-frequency oscillations could challenge system-wide stability, requiring advanced synchrophasor measurements. Additionally, forced oscillations from AI workloads, ranging from 0.5-1.5 Hz [5], overlap with inter-area oscillation modes. Research indicates that distributed data center deployments amplify these oscillations more than concentrated ones, contradicting conventional grid planning views. A comprehensive classification of these oscillatory phenomena, including frequency ranges, origins, detection, and mitigation methods, is provided in **Table III**.

#### C. Voltage Sensitivity and Protection Mismatch

The voltage sensitivity of data centers presents a significant vulnerability, influenced by various factors. The ITIC curve indicates disconnection at 0.7-0.8 p.u. voltage for over 20

milliseconds [19][25], aimed at protecting IT equipment, but implementation varies widely among facilities. ERCOT's data shows that identical voltage disturbances can lead to load reductions between 39% and 100% across different setups [26]. Differences in UPS technology, such as automatic reconnection in static systems versus manual intervention in Dynamic/Diesel Rotary UPS (DRUPS) systems, contribute to this inconsistency [15][27]. Additionally, variations in protection philosophy and firmware configurations can lead to further discrepancies [18][24]. The interaction between utility operations and data center protections creates vulnerabilities; for instance, utilities may attempt three automatic reclose operations for temporary faults. Some data centers count these as separate disturbances, resulting in disconnecting after the third attempt, even if disturbances are within tolerance limits. This miscoordination led to a significant 1500 MW disconnection event, highlighting the systemic risks from independently developed protection philosophies [6][19].

#### D. IBR Integration Compound Effects

The interaction between AI data centers and IBR-dominated power grids presents significant stability challenges. Al Kez and Foley note that systems with approximately 70% IBR penetration can experience frequency collapse due to unpredictable AI data center load changes [3], whereas synchronous systems manage these ramps effectively.

The lack of inherent inertia in IBRs reduces synchronous inertia, thereby decreasing the system's capacity to handle power imbalances caused by large AI loads. This results in higher rates of change of frequency (RoCoF) during load transitions, approaching protection system limits. Additionally, limited fault-current contributions from IBRs complicate protection coordination and heighten voltage-sensitivity issues [8], [28].

Control dynamics between remaining synchronous generators and IBR controls can lead to complex interactions, with the potential for undamped oscillations from grid-following inverters [7], [17]. As IBR penetration increases significantly, critical grid stability metrics for AI operations degrade, with trends including a reduced frequency nadir, extended voltage recovery times, and a shift in oscillation damping from positive to negative at higher penetration levels. These factors raise concerns about system instability, particularly when combined with variable AI workloads [5][21].

TABLE III. OSCILLATORY PHENOMENON CLASSIFICATION<sup>3</sup>

| Ref. | Frequency Range | Origin Mechanism             | Detection Method       | Mitigation Approach   |
|------|-----------------|------------------------------|------------------------|-----------------------|
| [5]  | 0.5 – 1.5 Hz    | AI training cycles           | Standard PMU           | Workload Scheduling   |
| [7]  | 10.0–11.0 Hz    | UPS control instability      | High rate PMU          | Firmware updates      |
| [7]  | 14.7–14.8 Hz    | Limit cycle from instability | Specialized monitoring | Controller retuning   |
| [24] | 23 Hz           | PE interaction               | DFR required           | Component replacement |

<sup>3</sup> UPS is the acronym of uninterruptible power supply.

#### IV. MITIGATION STRATEGIES: TECHNICAL AND INSTITUTIONAL DIMENSIONS

This section comprises 1) hierarchical control and storage architecture, 2) advanced grid codes and interconnection requirements, and 3) operational coordination paradigm.

#### A. Hierarchical Control and Storage Architecture

The multi-timescale nature of challenges necessitates hierarchical mitigation approaches. The investigation revealed successful strategies that employ different technologies, which are matched to specific timescales:

**Microsecond-Second Response:** Supercapacitors and power electronics modifications address switching transients and brief power quality events. Google’s compiler-based power shaping [4] operates at this scale, redistributing computational operations to smooth power profiles. Additionally, Google found that simple CPU utilization metrics can predict data center power consumption within 5% accuracy for over 95% of their power distribution units, despite complex AI hardware mixes [38]. It redistributes computational operations to smooth power profiles with less than 1% performance impact. This approach seems promising.

**Second-Minute Response:** Battery energy storage systems appear to be effective here. Systems sized at 350-450 MVA demonstrate effectiveness for frequency regulation during AI load ramps [3]. Tesla’s Megapack deployments show a 70% reduction in power variability when properly controlled [20].

**Minute-Hour Response:** Thermal energy storage leverages cooling system thermal mass for load shifting. Zhang et al. [22] demonstrate that 600 m3 cold water storage enables energy arbitrage while maintaining temperature requirements. This is a clever use of existing infrastructure.

The interaction between storage layers requires coordinated control, something often overlooked. Li and Li [17] emphasize that bandwidth limitations of upstream converters constrain overall system response. Controlling architecture must respect these fundamental limits.

#### B. Advanced Grid Codes and Interconnection Requirements

Regulatory frameworks are slowly evolving to address AI data center characteristics. Alberta’s AESO requirements [8] represent the first of their kind and the most comprehensive regulatory framework available. It clearly set out the Oscillation limits (<16 kW variation per 100 milliseconds), Ramping constraints ( $\leq 10$  MW/minute), Ride-through obligations (0.15 seconds at <0.45 p.u. voltage), and the Most Severe Demand Contingency (MSDC) compliance for all configurations. ERCOT’s Advanced Remedial Action Scheme [29] takes a different approach. It introduces performance-based capacity allocation. Facilities demonstrating 20-cycle (333 millisecond) response capability gain access to full grid capacity. Slower responses receive proportionally reduced allocations. This creates economic incentives for flexibility investment, a market-based solution that appears to be working. The analysis presented in [37] indicates Virtual Power Plants could provide 40-60% cost savings compared to conventional alternatives. **Table IV** compares the regulatory frameworks currently being developed by different utilities, highlighting the variation in requirements, enforcement mechanisms, and observed effectiveness in addressing AI data center integration challenges.

#### C. Operational Coordination Paradigm

The U.S. Department of Energy’s “shared energy economy” concept [30] represents something new, a fundamental reconceptualization of data center-grid relationships. Rather

than treating data centers as passive loads, this framework positions them as active grid participants. This setup could further provide 1) ancillary services using backup generation and storage for frequency regulation & reactive power support, 2) demand response leveraging computational flexibility for load management, and 3) infrastructure investment through co-funding transmission and generation projects. Implementation faces both technical and institutional barriers. Technical challenges include real-time coordination between computational scheduling and grid operations. This is not trivial. Institutional barriers involve regulatory frameworks designed for unidirectional power flow and fixed load profiles.

TABLE IV. REGULATORY FRAMEWORK COMPARISON <sup>4</sup>

| Ref.            | Utility | Key Requirements                                              | Enforcement Mechanism  | Effectiveness                  |
|-----------------|---------|---------------------------------------------------------------|------------------------|--------------------------------|
| [8]             | AESO    | Quantified limits for oscillations, ramping, and ride through | Mandatory compliance   | High specific thresholds       |
| [29]            | ERCOT   | Performance-based capacity allocation                         | Economic incentives    | Medium voluntary participation |
| [1], [27], [28] | NERC    | Risk framework and guidelines                                 | Industry collaboration | developing                     |

<sup>4</sup> AESO, ERCOT, and NERC are the acronyms of Alberta Electric System Operator, Electric Reliability Council of Texas, and North American Electric Reliability Corporation, respectively.

#### V. SYNTHESIS AND FUTURE DIRECTIONS

This section discusses the synthesis and future direction with 1) an interconnected challenge framework, and 2) respective research priorities.

##### A. Interconnected Challenge Framework

The analysis highlighted that AI data center challenges are interconnected across multiple time scales. Issues at the microsecond scale, such as power electronics constraints, affect millisecond protection system operations, which in turn lead to second-scale ramping challenges and hour-scale thermal cycling. Solutions must be integrated across these scales; for example, compiler-based power shaping addresses microsecond fluctuations but requires coordination with battery and thermal storage systems [4]. Regulatory frameworks need to account for these interactions while ensuring reliability.

**Fig. 2** shows how microsecond power electronics switch cascades through various scales—from millisecond protection operations to hour-scale training patterns. This cascading effect illustrates the need for coordinated multi-timescale approaches, as traditional single-scale solutions are inadequate. Furthermore, the spectrum of AI data center oscillations (0.1-23 Hz) overlaps with AI workload forcing functions and traditional inter-area oscillations, emphasizing the need for upgraded monitoring infrastructure and new frameworks for stability assessment.

##### B. Research Priorities

The analysis found critical knowledge gaps not covered in the literature, particularly regarding the simultaneous operation of multiple AI data centers on standard transmission systems and their systemic effects. Koomey’s analysis revealed that national projections can obscure significant regional variations, with some utility forecasts regularly surpassing actual generation [36]. The potential for interference patterns remains uncertain, and the scalability of potential mitigation approaches requires validation, necessitating testing of solutions demonstrated at single facilities at the system scale.

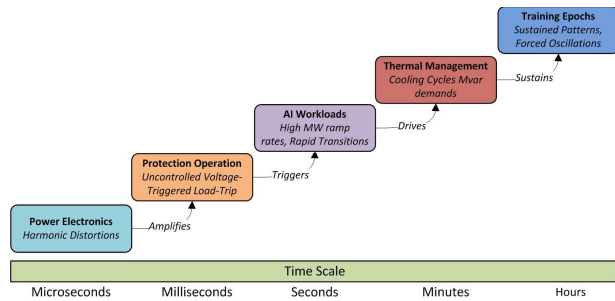


Figure 2. AI data centers multi-scale challenge cascade.

## VI. CONCLUSION

This proposed review highlights that AI data centers signify a significant shift in power system loads, going beyond mere growth. Their unique characteristics and integration with inverter-based resources (IBR) present challenges that current infrastructure and practices struggle to address.

Key takeaways include the need for hierarchical solutions due to the multi-timescale nature of these challenges; no single technology can provide complete mitigation. The shift to IBR-dominated grids will require a fundamental reevaluation of solutions originally designed for synchronous systems.

To successfully integrate AI data centers as proactive grid participants, technical innovations, regulatory changes, and new market mechanisms are essential. Future power systems must factor in AI workload characteristics and develop new modeling frameworks tailored to these demands, alongside specific technical standards for computational loads. Collaboration among utilities, data center operators, and researchers will be crucial in tackling these significant challenges.

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